

The ZigZag Premium: Does Career Diversification Produce Measurable Results For Liberal Arts Alumni?

A study of Macalester College Alumni

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ECON 354 Term Paper

1. Introduction

Career advice for decades has been fundamental: pick a lane, specialize, and continue climbing the ladder. One's long-term commitment to a field signals expertise and therefore success in future endeavors. However, in the recent Artificial Intelligence (AI) boom, we notice a growing demand for “thinkers” over individuals who are specialized. Through this paper, we attempt to dive into the psyche of an economy that is new, and we hypothesize that this market favors the ability to think across problems given AI abstracts much of the technical know-how.

As AI begins to handle routine cognitive labor, the roles that are most likely to get reshaped are the ones that anchor these linear paths. Notice how we use the term *reshape* and not *replace*. BCG's 2026 workforce study is explicit on this point: of the 43% of US roles where AI automation potential crosses a meaningful threshold, the vast majority will be augmented rather than replaced meaning the human stays, but everything around them changes. They estimate that 50% to 55% of all US jobs will be reshaped within two to three years, while full elimination touches only 10% to 15% over a longer horizon. *The specialist is not being fired. They are just being asked to do something they were never traditionally trained for.*

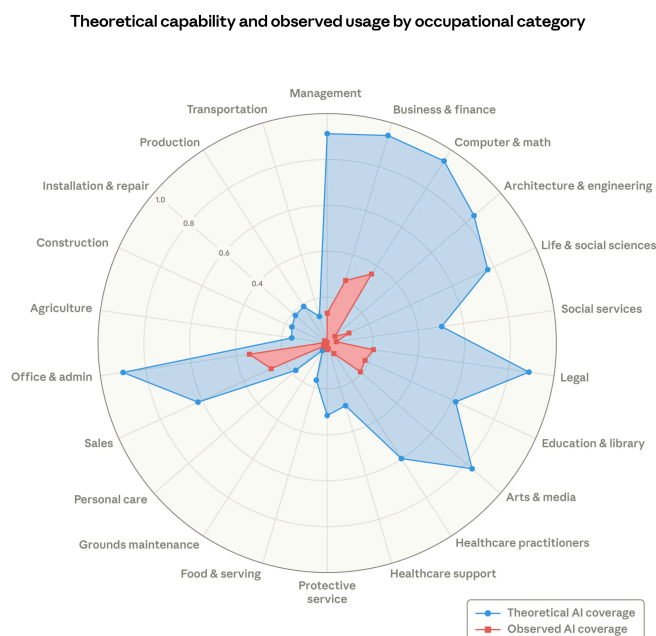


Figure 1: Theoretical vs. Observed AI Capability by Occupation

The graph above from Anthropic's research is a good reminder for the rapidly growing reach and ever-evolving nature of AI models. The blue in that chart is where AI could go. The red is where it already is. The distance between them is not a permanent ceiling, it is the adoption gap closing in real time. Notably, the widest gaps show up in Business & Finance, Management, and

Computer & Math. These are the same occupational categories that anchor the career track for a typical Macalester STEM major. The Anthropic researchers found that the workers most exposed to AI are not low-skill, rather they are students like us. *The ZigZag premium we find in our data may, in part, be an early hedge against exactly that labor market pressure.*

2. Hypothesis

We test four claims through this paper. Our primary hypothesis is that alumni with higher diversification scores, having pivoted across industries and functions, are positioned for more career success - when controlling for confounders like age and graduate degrees. We explore the following mechanisms and interactions in our data:

- a. **Velocity** :- Diversification does not just predict where you end up, but also *how fast* you get there.
- b. **Grad School** :- Having accounted for actual “on-the-ground” experience and transitions between roles, we suspect graduate degrees play a smaller role on seniority than commonly believed.
- c. **Neutrality** :- Does this claim hold across college majors in our dataset; we seek to explore whether or not one can pin down a broad mechanism beyond economics and computer science tracks.

Research on labor-market mobility suggests that career advancements often come from movement across roles. Tate and Yang show that breadth and varied experience strengthen internal labor-market outcomes and the Richmond Fed notes that job changes are part of proven wage growth. Together, we see evidence backing the idea that dynamic career paths are linked to long-term success.

We extend this argument by adding that “Zig-Zagging” through your career and gathering broad experiences is a strong indicator of career success. If true, a liberal arts education today is more valuable than ever.

3. Data Description & methodology

a. Data Collection:

The dataset contains 289 Individual Macalester alumni with information ranging from experience, education, and various other parameters collected from LinkedIn as of April 2026 to construct our variables (refer to Appendix Exhibit 1A). Sampling was done randomly. The data was then entered manually into an organized log. The raw profile data was stored in a long-form experience table (one row per position) and then aggregated at variable level. Scoring rubric is

uniform across all observations. Due to the potential for inaccurate results from incomplete alumni records, we had to exclude observations that were missing a graduation year or had an incomplete work history. After excluding these individuals, we had 202 complete observations for our study. Table 1 below presents a statistical summary of our variables, discussed contextually in the variable construction section:

Variable	N	Mean	SD	Min	Median	Max
Diversification Score (0–100)	202	62.02	23.72	10	63.5	100
Industry Count	202	2.86	1.45	1	3.0	7
Function Count	202	3.36	1.73	1	3.0	8
Career Moves Count	202	3.64	2.63	0	4.0	17
Seniority Score (0–5)	202	2.80	1.33	1	3.0	5
Career Velocity Score (0–100)	202	75.94	13.17	30	78.0	95
Leadership Signal Count	202	2.50	1.99	0	2.0	9
Career Success Composite (0–100)	202	75.54	13.99	25	78.0	96
Graduate Degree (0/1)	202	0.35	0.48	0	0.0	1
Years Since Graduation	202	10.24	10.26	1	7.0	54

Table 1: Descriptive Statistics of Macalester Alumni Dataset (N=202)

b. Variable Construction

To identify the relationship between diversification and success, the main independent variable for this study will be the **Career Diversification Score** (0 – 100). This score measures career breadth based upon all full-time post-graduation positions. While it would have been easy to simply count the total number of jobs taken by alumni, developing a multi-dimensional measure of job movement per individual adds additional depth to this research. The score includes two dimensions of job movement: industry-based (e.g., financial services, healthcare, tech, consulting, non-profit) and function-based (e.g. finance, operations, strategy, engineering, legal). By design, campus employment, preceptorships and internship opportunities prior to graduation were not included when calculating this score. For example, an alumnus who has worked in three different finance roles would receive a lower score using this measure than someone who has gone from working in asset management to product strategy to leading a non-profit organization. Based upon the current sample, the mean Diversification Score was 62.8 (Median: 65.0; StDev: 22.8); while the range of possible scores varied from 10 to 100.

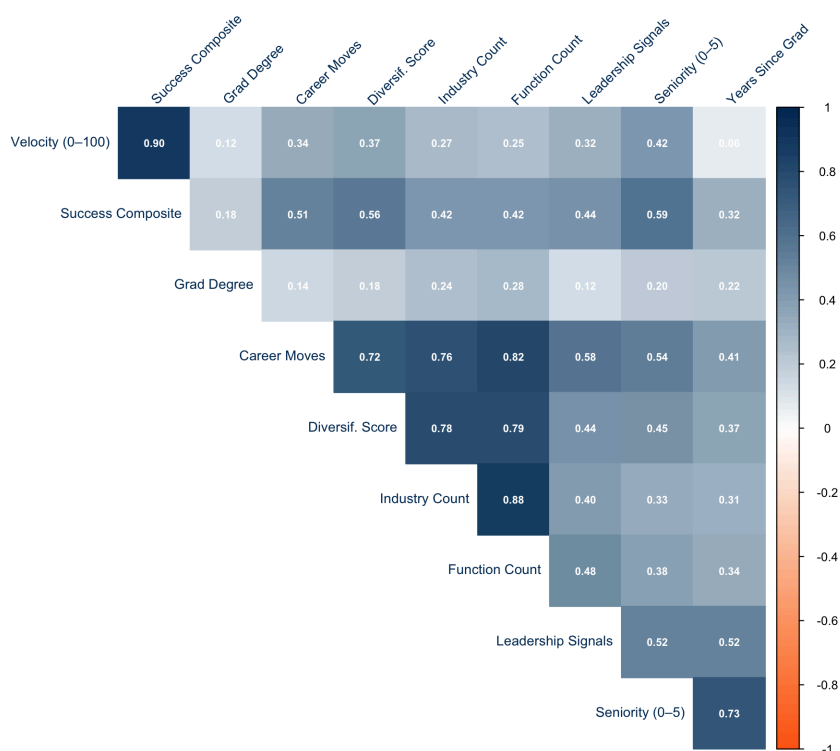


Figure 2: Heatmap of Career Metrics

The correlation matrix in Figure 2 reveals some internal dynamics of career success in our samples. Three key relationships stand out: 1. Velocity and Success Composite move almost together ($r = 0.90$), 2. Diversification and Success are strongly correlated ($r = 0.56$), and 3. Having a graduate degree has a weak correlation with Success ($r = 0.18$). The high velocity-success correlation shows that success today is defined by the speed of advancement over being at a role long term or waiting to get positions from seniority. Successful graduates aren't just moving through ranks quickly but also have more career pivots. Similarly, diversification is a stronger predictor of success than a graduate degree so 'learning by doing' is almost three times more valuable.

Two structural observations include a strong correlation between Industry Count and Function Count ($r = 0.88$), meaning almost always, on average, a cross sector pivot is also a cross functional change. When an alum changes their industry, they are almost always taking on a new functional role. An important challenge with our dataset was accounting for the seniority that older graduates naturally have, fortunately, a structural trait of our set up can be seen through a contrast between Seniority and Velocity against Years Since Graduation ($r = 0.73$ versus $r = 0.06$). This confirms that the velocity score is independent of the time effect from seniority. A 1995 graduate and a 2022 graduate are on a level playing field in the velocity regression since both are being measured against how fast their own cohort climbed, not against each other.

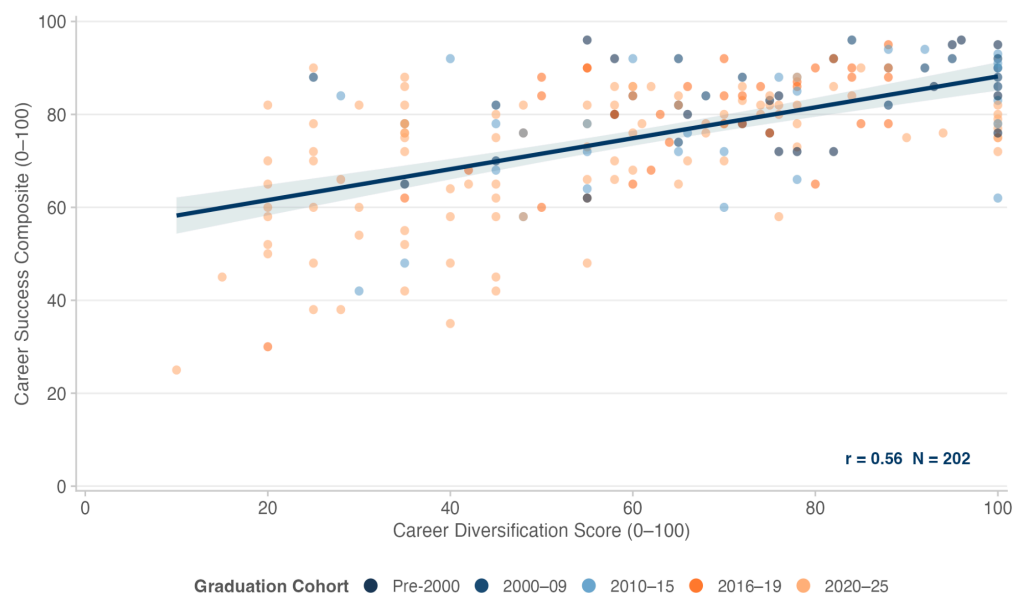


Figure 3: Scatterplot of The Success Gap

The scatter plot from Figure 3 bolsters our base case before a more thorough regression analysis. Alumni in the top diversification tier outperform the bottom tier by 17 points which represents an entire standard deviation in our success measurement. What's particularly striking from the perspective of Figure 3 is that regardless of cohort there appears to be consistency within the steepness of the slope. The "zigzag premium" does not appear to be either a product of the current time period or some anomaly related to a previous time period but rather what appears to be a structure inherent to the nature of the Macalester career path.

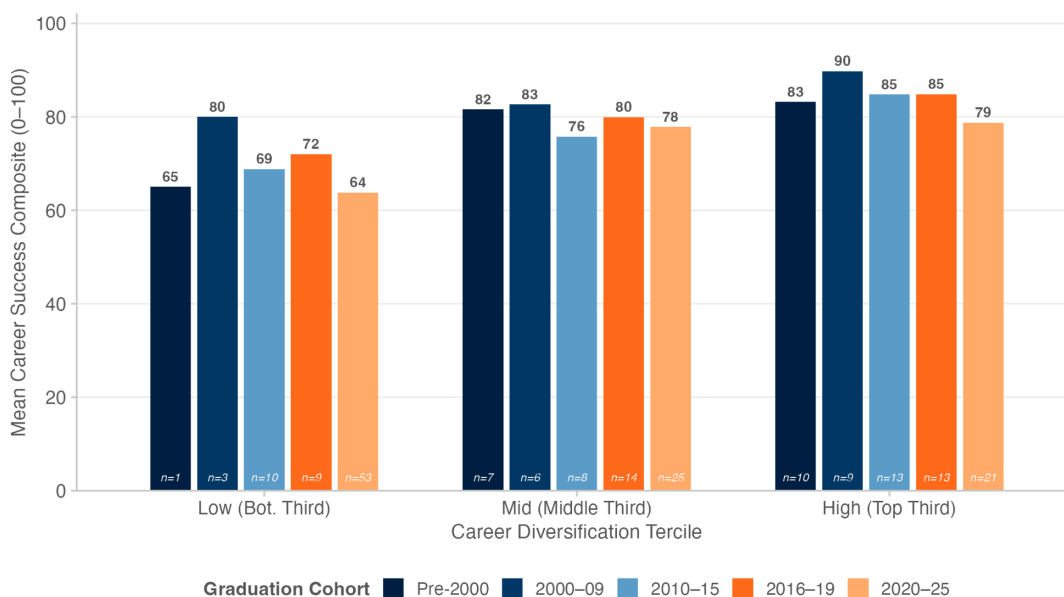


Figure 4: Barplots of ZigZag Premium Across Time and Diversification

Figure 4 answers a natural follow up question: how has this relationship changed over time? Across the five graduation buckets, we see that the gap between top and bottom diversification terciles holds across all of them. For alumni who graduated before 2000, high diversifiers averaged 83 on the Success Composite against 65 for low diversifiers, an 18-point spread. For the 2020–2025 cohort, the numbers are 79 and 64, a 15-point gap. This is the largest difference seen across all recent graduates, this was also our largest cohort ($n = 63$). While it does represent new entrants into the workforce and while they have fewer opportunities to transition jobs or accumulate experience, there is still a clear advantage when they do.

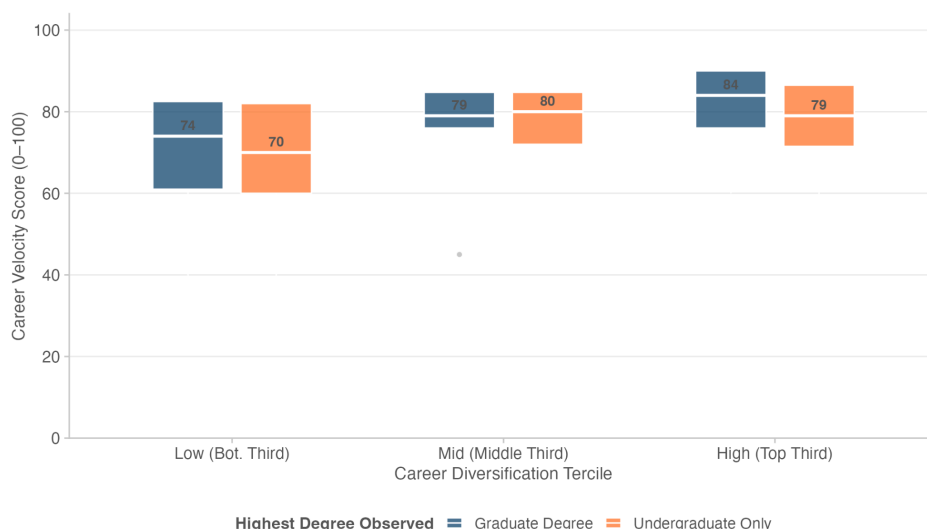


Figure 5: Boxplot of Career Velocity by Diversification and Degree Type

Figure 5 addresses the most common pushback against career pivots: that changing direction slashes career momentum. Among alumni with graduate degrees, median velocity rises from 74 points in the bottom diversification pool to 84 in the higher diversification pool. Alums with only undergraduate degrees also see a jump from 70 to 79. What is most striking is the marginal return on the pivot vs. the degree: the jump in career velocity attributed to a graduate degree is smaller (4 to 5 points) than the diversification effect (9 to 10 points) at any tercile. Highly diversified career paths tend to cluster more around the median, low diversifiers are more spread out - it seems, broadly, diversified careers seem to sit at similar career velocity ranges.

Case Study on early career ZigZag premium: Jackson Tak's career progression (Graduating Class of 2021), demonstrates clearly how the Zigzag premium works during an individual's first few years on the job. Tak's Diversification Score of 45 places him below the median which indicates he has been in the workforce for less than one decade to establish multiple cross-industry moves. However, his Career Velocity Score of 70 acts as a proxy to account for time in our model. His career progression took him from military service into boutique investment banking, and then into corporate development at Sezzle in less than five years. Tak was able to get a "Velocity Premium", typically earned by those with greater experience than he has, by diversifying early. His track record indicates that while it is possible to measure diversification by the number of times an individual crosses an industry boundary, measuring its value or impact is most relevantly tied to both timing and the magnitude of each transition. While changing careers may help a recent college graduate to achieve increased career velocity, a single well-timed move from banking to a high growth fintech such as Sezzle, can provide similar value. That is why we have created our model to be multi-dimensional, we should account for both the strategic leap-frogging benefits associated with making a successful pivot early in one's career versus simply rewarding longevity.

5. Regression Analysis

a. Regression Mechanism

To test our hypothesis, we used an OLS regression model with the Career Success Composite as the dependent variable. Five model specifications are estimated sequentially, adding controls incrementally to test robustness. Model 1 regresses Success on Diversification Score alone. Subsequent models add Years Since Graduation, Graduate Degree, Leadership Signal Count, and Major Type in sequence.

b. Regression results

	Success Composite (0–100)			Velocity (0–100)	Seniority (0–5)
	Base	Add	Full	H2	H3
Diversification Score	0.305*** (0.037)	0.263*** (0.038)	0.271*** (0.037)	0.222*** (0.039)	0.009*** (0.003)
Graduate Degree (0/1)	—	2.129 (1.705)	2.015 (1.699)	1.980 (1.845)	0.083 (0.132)
Leadership Signal Count	—	1.592*** (0.489)	1.403*** (0.485)	—	0.089** (0.038)
Economics Major (0/1)	—	—	–0.305 (2.746)	0.495 (2.983)	—
STEM/Quant Major (0/1)	—	—	8.155** (3.703)	8.892** (4.001)	—
Years Since Graduation	0.170** (0.085)	0.024 (0.093)	0.086 (0.095)	–0.078 (0.093)	0.077*** (0.007)
Constant	54.864*** (2.270)	54.252*** (2.228)	46.554*** (3.415)	53.855*** (3.701)	1.197*** (0.173)
N	202	202	202	202	202
R²	0.331	0.370	0.396	0.191	0.580
Adjusted R²	0.325	0.357	0.378	0.171	0.572

*SEs in parentheses. *p<0.10; **p<0.05; ***p<0.01. HC3 robust SEs confirmed. Observational data — do not interpret as causal.*

Table 2: OLS Regression Results for Career Success and Velocity
Impact of Diversification, Credentials, and Major Type on Professional Attainment (N=202)

Looking at Table 2, the results support our hypothesis, and the regression is layered allowing us to identify drivers of successful outcomes for the Macalester graduate. While Years Since Graduation is a statistically significant factor for predicting success (as indicated by 0.170) in the basic model, when we include the Leadership Signal Count, this factor becomes statistically non-significant (0.024). Therefore, we see that time, by itself, is not a determinant of success; however, accumulating leadership experiences throughout the years is the actual driver of success. When we include Leadership Signal Count as a statistically significant (1.592) indicator of success, it indicates that the model effectively captures the professional value of early-career initiative.

When we consider the Diversification Score, it demonstrates remarkable resilience. The coefficient moves from a value of 0.305 in the basic model to 0.271 in the full model which includes Leadership Signal Count, graduate degrees, and various majors. Interestingly, Graduate Degree Attainment is not statistically significant in any of our models. Therefore, attaining another degree does little to enhance one's Success Composite once both diversification and leadership are considered. In addition, based upon our analysis of the data, we can conclude that the market has shifted its valuation criteria from academic credentials toward functional agility regarding the contemporary "thinker."

To test for velocity mechanism, we developed a parallel model (Table 2, Column H2) in which our cohort-adjusted velocity score is used as an outcome measure. Our data supports our hypotheses regarding velocity. For instance, we found diversification to be a significant predictor of the speed with which one climbs the ladder (0.285), after controlling for time-in-market. Therefore, while it may appear that 'zigzaggers' are able to reach higher levels on the ladder by virtue of having been around longer, they are actually moving up the rungs of the career ladder at a much greater pace than those who are focused on a single specialty.

Although the STEM/Quant Major variable creates a baseline advantage of approximately 8.2 points in success and 8.9 points in velocity, diversifying one's background serves as the great equalizer. Of particular note is the fact that there is no distinct "Economics Premium" associated with being an Economics major within this sample since the variable is not statistically significant. A person whose Diversification Score is 30 points greater than a peer with a specialized background will effectively earn approximately 8.1 points of predicted success ($30 \times 0.271 \approx 8.1$). Therefore, although there exists a STEM-related advantage within this sample, strategic diversification offers a clear pathway to closing the gap.

6. Conclusion

We began this project out of curiosity for how a smaller liberal arts school will shape us in what is seemingly the revolution of a century. In a four-year span, words like "Investment Banker", "Consulting", "Interview Cycle", and "Target School" define a good part of how you are and will continue to define success for yourself - a revolution that shakes the foundations of what capable candidates look like is relieving. That is not to say that schools like Macalester are not home to rigor that matches those of more specialized schools; rather, it is an inherent need for the average individual to be truly thoughtful and resourceful stewards of society. This need blooms from an underlying ethic that builds on a shared understanding that an appreciation for the other and resourceful, curious outlook wherever we may be will lead us to a way.

Our research shows that top diversifiers do not simply "do better" - rather they outperform the bottom tercile by a large margin. The linear career and learning systems built to feed into it will

likely be subject to significant reevaluation. This is exactly what Zia Yusuf, Sean Epp, and Ed Ray emphasized in our Deals class. Whether it's moving from distressed debt to dental rollups or from a parking startup to BCG, the most successful people in our network are the ones who treat their careers like strategic moves inspired by curiosity over traditional, defined success routes.

For students entering a highly unpredictable and dynamic environment, be open to a little “ZigZaging” as you maneuver through it the following decade.

References

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4. Tate, and Yang. *The Bright Side of Corporate Diversification: Evidence from Internal Labor Markets*.
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Appendix

1a.

Variable	What it captures	Type
Profile level (one entry per person)		
Full name	Identity link across sheets	Observed
LinkedIn headline	Current stated role or identity	Observed
Location	Current geography	Observed
Date collected / collector	Data provenance and consistency checks	Observed
Raw profile text	Full paste; source of truth for all coded variables	Observed
Experience (one row per role)		
Organization name	Employer or institution	Observed
Role title	Exact title as listed	Observed
Employment type	Full-time, part-time, internship, seasonal	Observed
Start / end date	Tenure dates; used to compute duration	Observed
Duration (months)	Calculated from start/end dates	Observed
Is current role (0/1)	Flags whether role is ongoing	Observed
Is internship (0/1)	Excludes from core career path; retained for controls	Observed
Is campus/student role (0/1)	Excludes preceptor/TA roles from diversification count	Observed
Include in core post-grad analysis (0/1)	Main career-path flag; excludes pre-grad roles	Coded

Industry bucket	Broad sector (e.g. Financial Services, Healthcare, Tech, Nonprofit)	Coded
Function bucket	Type of work (e.g. Finance, Strategy, Engineering, Legal, Teaching)	Coded
Role seniority score (0–5)	Level within organization; follows consistent title hierarchy	Coded
Leadership indicator (0/1)	Flags management, team leadership, or named club presidency	Coded
Career stage	Undergraduate / early career / mid career label	Coded
Education (one row per institution)		
Institution name	Degree-granting school	Observed
Degree type	BA, MBA, JD, MD, PhD, etc.	Observed
Field of study / major	Declared major(s) and minor(s)	Observed
Graduation year	Anchors the cohort control variable	Observed
Is Macalester (0/1)	Flags the undergraduate record used for grad-year control	Observed
Derived regression variables (one row per person)		
Career Diversification Score (0–100)	Weighted breadth of industry + function crossings across core post-grad roles	Derived
Career Success Composite (0–100)	Bundled outcome: role selectivity + employer prestige + leadership signals	Derived
Career Velocity Score (0–100)	Seniority attainment relative to graduation cohort; strips age effect	Derived
Highest seniority score (0–5)	Peak seniority level observed across all post-grad roles	Derived
Industry count	Number of distinct industry buckets crossed	Derived
Function count	Number of distinct function buckets crossed	Derived
Career moves count	Number of employer transitions in core post-grad path	Derived
Leadership signal count	Total observable leadership indicators across post-grad roles	Derived
Graduate degree (binary)	Presence of any post-BA degree (MBA, JD, MD, PhD, etc.)	Derived
Major bucket (categorical)	Consolidated field grouping for major neutrality test	Derived
Years since graduation	2026 minus Macalester grad year; main cohort control	Derived